

Algorithm and Model Changes

magnetic-field-inspired-navigation

2026-05-22

This document records major algorithm and model decisions in the cleanroom `magnetic-field-inspired-navigation` implementation.

Rule for this repo

- Any major change to the controller, dynamics model, obstacle model, sensing abstraction, or mathematical formulation must be recorded in this file in the same change that introduces it.

The purpose is to keep a durable map between:

- the legacy ROS implementation in `double_integrator`
- the cleanroom Python implementation
- intentional simplifications or deviations

Entry 2026-05-22: Initial Cleanroom Double-Integrator Reference

Status

- Implemented in `src/mfinav/double_integrator.py`
- Scenario runner in `scripts/run_reference.py`

Legacy source anchors

- Goal controller and relaxation: `nodes/ataka_listener.py`
- Duplicate goal controller: `nodes/multi_agent_listener.py`
- Magnetic avoidance, `field == 2`: `nodes/collision_avoidance_okt.py`
- Double-integrator rollout: `nodes/system_motion.py`

What Was Implemented

The current cleanroom reference includes:

- 2D double-integrator dynamics
- 2D circular obstacle abstraction
- goal-relaxation gain logic inspired by the ROS implementation
- magnetic-field-inspired obstacle avoidance based on the ROS `field=2` branch

- additive total command

$$a = a_{\text{goal}} + a_{\text{obs}} \quad (1)$$

$$v_{k+1} = v_k + a_k \Delta t \quad (2)$$

$$p_{k+1} = p_k + v_{k+1} \Delta t \quad (3)$$

Goal Relaxation Implemented

Let

- $p \in \mathbb{R}^2$ be robot position
- $v \in \mathbb{R}^2$ be robot velocity
- $g \in \mathbb{R}^2$ be goal position
- $r_g = g - p$ be the goal vector
- r_o be the closest obstacle vector
- $d_g = \|r_g\|$
- $d_o = \|r_o\|$

The relaxation scale used in the cleanroom implementation is

$$k_{\text{relax}} = \left(1 - e^{-d_o/1.5}\right) \cdot w_{\angle} \cdot w_{\text{prog}} \cdot w_{\text{exit}} \quad (4)$$

where

$$w_{\angle} = 1 - \cos \theta = 1 - \frac{r_g^{\top} r_o}{\|r_g\| \|r_o\|} \quad (5)$$

$$w_{\text{prog}} = \begin{cases} 1, & d_g \leq d_{g,0} \\ \exp\left(-\frac{d_g - d_{g,0}}{0.1}\right), & d_g > d_{g,0} \end{cases} \quad (6)$$

and

$$w_{\text{exit}} = \exp\left(-\frac{d_g - d_{g,\min}}{0.1}\right) \quad (7)$$

with:

- $d_{g,0}$: initial distance to the goal
- $d_{g,\min}$: smallest distance to the goal reached so far

This follows the structure of the ROS code around the `goal_relax == 1` block.

Goal Controller Implemented

For the far field, the current cleanroom implementation uses a simplified speed-and-turn controller inspired by the ROS geometric block:

$$a_{\text{goal}} = a_{\text{attr}} + a_{\text{turn}} \quad (8)$$

$$a_{\text{attr}} = -(k_p k_{\text{relax}}) (\|v\| - v_{\text{max}}) \hat{s} \quad (9)$$

where $\hat{s}$ is:

- the normalized velocity direction if $\|v\| > 0$
- otherwise the normalized goal direction

The turning term is

$$a_{\text{turn}} = k_{\omega} k_{\text{relax}} \|v\| \hat{n} (\hat{n}^{\top} \hat{g}) \quad (10)$$

where:

- $\hat{g} = r_g / \|r_g\|$
- $\hat{n}$ is the left-perpendicular of the current heading

Near the goal, the controller switches to a PD-like attractive form:

$$a_{\text{goal}} = k_{p,\text{goal}} r_g - k_d v \quad (11)$$

Magnetic-Field-Inspired Obstacle Term Implemented

The cleanroom port uses the `field == 2` structure from the old code.

Let

- r_o be the closest vector from robot to obstacle surface
- $d = \|r_o\|$
- $\hat{l}_a = v / \|v\|$ be the normalized robot motion direction
- $r_{\text{from obs}} = -r_o$
- $\hat{r} = r_{\text{from obs}} / \|r_{\text{from obs}}\|$

The obstacle current direction is the tangential projection of the robot motion:

$$\hat{t} = \hat{l}_a - (\hat{l}_a^{\top} \hat{r}) \hat{r} \quad (12)$$

If this becomes degenerate, the implementation falls back to a perpendicular direction.

The 2D magnetic field surrogate is modeled as

$$B \propto \frac{\|v\|}{d} \hat{t}_\perp \quad (13)$$

and the magnetic avoidance term is

$$F_{\text{mag}} = k_m \hat{l}_{a,\perp} (\hat{l}_{a,\perp}^\top B) \quad (14)$$

This is a 2D analogue of the nested cross-product structure in the ROS code:

$$B = [\hat{t}]_\times v / d \quad (15)$$

$$F = k_m [\hat{l}_a]_\times B \quad (16)$$

Added Repulsive Correction Implemented

The cleanroom implementation also keeps the old additional repulsive term near the obstacle:

$$F_{\text{rep}} = -k_r \left(\frac{1}{d} - \frac{1}{d_b} \right) \frac{r_o}{d} \quad \text{for } d \leq d_b \quad (17)$$

Then it is projected to remove the component along the motion direction:

$$F_{\text{rep}}^\perp = F_{\text{rep}} - (F_{\text{rep}}^\top \hat{l}_a) \hat{l}_a \quad (18)$$

The final obstacle term is:

$$a_{\text{obs}} = F_{\text{mag}} + F_{\text{rep}}^\perp \quad (19)$$

Major Simplifications Relative To The Old ROS Code

These are intentional and important:

1. The current reference is strictly 2D. The ROS code contains both 2D and 3D branches.
2. Obstacles are currently circles. The ROS stack computes obstacle geometry from point clouds, polygonal segments, and local obstacle surfaces.
3. The far-field goal controller is structurally inspired by the ROS code, but it is not yet a line-by-line reproduction of the SO(3)-style geometric block.
4. The magnetic term is mathematically faithful in structure, but the current implementation uses a compact 2D analogue rather than a direct symbolic port of the original 3D cross-product expressions.
5. The cleanroom model removes all ROS-specific concerns: TF, publishers/subscribers, launch files, point-cloud nodes, timestamps, and platform adapters.

Why This Change Was Made

The goal of this first cleanroom step is verification by isolation:

- keep the algorithm small enough to reason about
- remove ROS coupling
- make rollout behavior inspectable with CSV and plot outputs
- establish a stable place for future refactors and comparisons

Required Follow-Up

Future major changes should update this file, especially if they modify:

- the exact goal-relaxation gain
- the magnetic obstacle formula
- the near-goal switching logic
- the state update model
- the obstacle abstraction
- the benchmark scenario assumptions